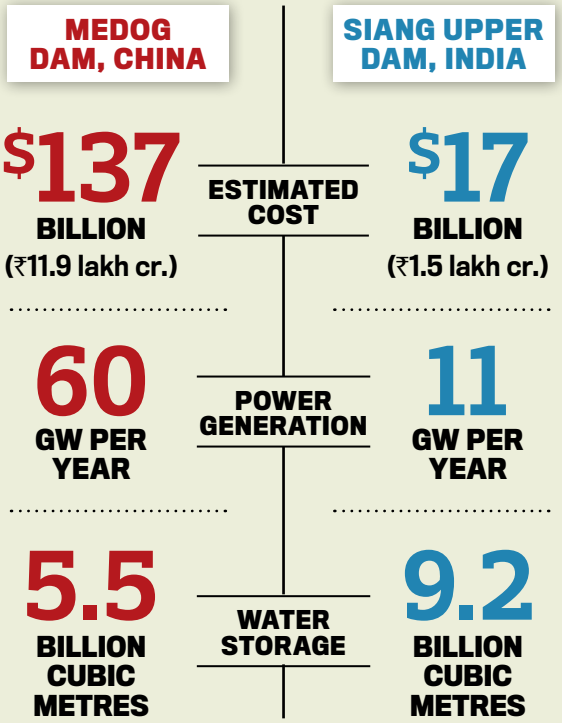
# CHINA'S MEGA DAM, INDIA'S COUNTER

**China is planning a massive hydropower project in Medog County, just 30 km from Arunachal Pradesh. In response, India aims to fast-track the Siang Upper Multipurpose Project at Parong in the state**




**Originating from the** icy grip of the Angsi Glacier, near the northern slopes of the Himalayas, close to Lake Manasarovar in western Tibet, the Yarlung Tsangpo—known as Siang in Arunachal Pradesh and the Brahmaputra in Assam—embarks on a long and winding 2,900 km journey. It flows eastward across the Tibetan Plateau, a seemingly calm giant, until it reaches the easternmost edge of the Himalayas. And then, it encounters an obstacle—a towering 7,782-metre peak called the Namcha Barwa. But the river does not stop. In one of nature's most breathtaking feats, it takes a dramatic U-turn around the mountain—the Great Bend, one of the sharpest and most spectacular river bends on Earth.

What follows is pure fury. The river plunges into the Yarlung Tsangpo Grand Canyon, a monstrous 500km-long chasm with depths exceeding 5,000 metres—nearly five times the height of Dubai's Burj Khalifa, the tallest structure in the world. It is one of the deepest and most treacherous canyons on the planet, where the river transforms into a roaring beast, its waters churning with an unstoppable force. This is where China wants to step in. The steep gradient at the Great Bend holds unparalleled hydroelectric potential that Beijing is determined to exploit. The plan? A colossal hydropower project embedded deep into the unforgiving terrain of Medog County, Tibet—one of the last unexplored and most geologically unstable regions on Earth. It is a project of staggering proportions, a testament to human ambition and engineering audacity.

With an estimated cost of $137 billion (Rs 11.9 lakh crore), this is poised to become the most powerful hydropower facility ever attempted, generating a jaw-dropping 60 GW of electricity annually—three times the output of China's own Three Gorges Dam, the current world record-holder, and surpassing the UK's entire annual energy consumption. Expected to be completed by 2033, the scale of the project is mind-boggling. To divert nearly half of the river's flow, Chinese engineers plan to drill tunnels up to 12.5 miles long through the Namcha Barwa mountain, rerouting 2,000 cubic metres of water per second—enough to fill three Olympic-sized swimming pools every second.

## India's Great Worry

While China presents the project as a step toward achieving carbon neutrality by 2060, for India and Bangladesh, it signals a looming disaster. The dam's location, barely 30 km from Arunachal Pradesh, places it alarmingly close to India's border, raising security concerns in New Delhi. Worst-case scenarios throw up a nightmarish picture: if the dam were to fail—due to engineering flaws, an earthquake or even sabotage—the consequences would be catastrophic. A towering surge of water could rip through Arunachal Pradesh and Assam, wiping out entire towns within minutes. "This Chinese project isn't just a tease," says Michael Kugelman, director of the Washington-based South Asia Institute. "China has the capacity to mobilise capital quickly and build out large-scale infrastructure projects without delays, thanks to its centralised economy and undemocratic politics. This means New Delhi needs to start thinking now about how to mitigate the possible deleterious implications—especially for water security, the environment and geopolitics."

The Brahmaputra is a vital artery that sustains millions across four countries—China (50.5 per cent), India (33.6), Bangladesh (8.1) and Bhutan (7.8). More than just a waterway, it sustains agriculture, drinking water and energy needs, making any disruption to its flow a serious concern. For India, the primary worry is China's ability to control the Brahmaputra's flow. If Beijing releases excess water during monsoons, devastating floods could ravage India's northeastern states, particularly Assam and Arunachal Pradesh, where nearly 40 per cent of the land is already flood-prone. The fallout—mass displacement, infrastructure collapse and economic devastation—would be severe.

Conversely, restricting water flow during dry months could cripple agriculture, hydropower generation and drinking water supplies. Given that the Brahmaputra accounts for nearly 30

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